

Discoveries

Original findings from a grown spiking creature, with no external antecedent

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The aibaby project — a C++17 spiking neural creature grown from a DNA/genome file

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Abstract

This document collects the findings of the aibaby project that are not held up by any published result. The project grows a spiking neural “creature” from a genome file and teaches it to produce vowel formants; much of its machinery is named after the literature it came from, and those mechanisms are deliberately excluded here. What remains is the set of things the system itself was the first to say: measured quantities with no external reference, structural obstacles that only appear in this architecture, and a group of inference failures that were found by being committed and then caught.

The methodological findings are listed first and at length, because they are the most transferable part and the least specific to a creature made of spikes. Several of them are reports of a mistake that survived review — a test no outcome could pass, a refusal criterion computed from the wrong arm, a statistic diagnosed in a comment and then quoted for months. They are included on the view that a failure mode with a worked example is more useful than a rule without one.

Every quantity below comes from a run gated on a pinned determinism hash, `ad96f882becbee92`, under which every mechanism that ships off is bit-identical. Section 8 gives provenance.

1. Scope, and what is excluded

A finding is included if the project discovered it, and excluded if a published result says it first or frames it. The line is drawn at the *claim*, not at the tools: the estimators used throughout are Cochran’s ratio estimator and the Quenouille–Tukey jackknife, and the quantities they produce are still this project’s, so they appear here while the methods do not.

Excluded on that basis are: the account of interference as failed contextual inference (Heald, Lengyel and Wolpert’s COIN model), the finding that context gating alone is insufficient without synaptic stabilisation (Masse, Grant and Freedman), importance-weighted protection of earlier weights (Kirkpatrick et al.; Zenke, Poole and Ganguli), node perturbation as a motor learning rule (Fiete, Fee and Seung), the stability–plasticity dilemma (Grossberg), and the uniform cluster prior implicit in MacQueen’s 1/wins rule. Each is cited in the project’s main reference list. Appendix A lists the project results that were excluded by this criterion, so the exclusion can be audited rather than taken on trust.

2. Measurement and inference

These came out of running the same experiment many times and being wrong in a way that printed a

number. None has an external source; each has a worked example in the project history.

2.1 Chance has no fixed value when the readout is “nearest”

A k-alternative naming score read against $1/k$ is not a chance comparison. When the decision rule is nearest-target in formant space, chance depends on the *geometry* of the targets: the same uninformative voice scores 0.161 against targets on the vowel corners and 0.306 against a 2×2 arrangement. An informationless voice reads about 0.30 on a four-word task where $1/k$ would predict 0.25. One earlier ceiling claim was a null error of exactly this kind: a real effect scored against an arithmetic chance level while the matched control sat in the same table, unread.

2.2 A test that nearly all outcomes pass is not a test, and neither is one that none can pass

Two symmetric failures, both found after they had produced a published-looking verdict. A guard written as $cd2 > -3 \cdot 13.1$ — “greater than -39.3 ” — passes on nearly any data, and was reported as “F2 is spared” when it meant “F2 is unmeasured”. In the other direction, an intervention’s success branch required a shift of 0.090 where a complete closure of the gap could produce only 0.070; it printed a confident conclusion from an unreachable branch. The fix in both cases is to print the power of the test beside its result, so that a criterion nothing could have cleared is visible rather than silent.

2.3 The reference point is the hypothesis’s own prediction, not zero

A hypothesis that names a value must be tested against that value. A drift account predicting a decay of -0.034 was first tested for being distinguishable from *zero*, which it was not, and so survived; tested against its own prediction, the measurement sat 3.9 SE away and the account was excluded. The same error in another guise is scoring an effect against $1/k$ while the real control is in the table.

2.4 A null and a reversal are different results

An *else* branch that catches both reports the wrong one. An intervention expected to close a gap was tested only for closure; when the gap *widened*, the outcome fell through to the null branch and was written up as “no effect”. The widening was 2.5 SE in the opposite direction to the prediction, which is a stronger result than the null it was reported as.

2.5 A constant copied instead of derived fails silently, and travels

Six instances in one 34,000-line file, in two families. *Indices*: a literal array size that was a copy of a named bound and overran into the next field; positional grid indices; positional arm indices that would have mislabelled whole rows; and pair indices ($k \cdot 2$, $k \cdot 2 + 1$) that silently shifted when two table rows were inserted above them. *Labels*: a band value copied beside the table it came from, so an experiment reported its widest condition as a tenth of its real width after the table was rescaled; and a deleted control whose label and refusal criterion were inherited by unrelated rows.

What makes this class survive review is specific and worth stating: **every site carried a comment acknowledging the indexing was positional, and the comment was treated as the fix**. One even warned about a *skipped* row reading zero — a different hazard from a *shifted* row, against which the guard written

for the first does nothing. A mislabelled number still prints with a standard error and still reads as a result, so nothing in the output looks wrong. The question that finds these is: *is this label attached to the data, or merely next to it?*

2.6 A passing determinism hash does not say the code is correct

The project gates every run on a pinned hash. A pass says the shipped configuration is unchanged and the run is reproducible; it says nothing about whether the experiment computes what its name claims. Four of the six errors in §2.5 were introduced and committed under a passing hash.

2.7 Reproduce the conditions, not the experiment name

A headline was called irreproducible after three runs at the registered default tick count. The recorded result had been taken at 3.5× that budget, which was written in its own log’s filename and in the note that cited it. A registered default is not evidence about what a recorded run used, and this system has a documented power curve across exactly that range — one experiment reads +1.0 at 560k ticks against +18.3 at 3.4M.

2.8 Diagnosing a broken statistic does not retire it

A retention measure was computed as a mean of per-creature ratios whose denominator — how much that creature learned — was floored only at $1e-6$. The code carried a comment beside it recording that the statistic had read -5.587 ± 6.729 and 27.222 ± 21.160 on another genome and calling it “a broken statistic reporting nothing”, and a denominator-free replacement was added alongside. The headline, the milestone gate and the pre-registered specificity criterion all went on reading the broken one.

Measured on the same 128 sessions, the two estimators disagree in *sign*: a mean of ratios gives $+0.182 \pm 0.663$ and a jackknifed ratio of sums gives -0.211 ± 0.074 , with standard errors differing ninefold. The transferable part is not the statistic but the order of operations: a comment records a fact, it does not change what the gates compute, and every consumer has to be moved by hand. Grep for the variable, not for the comment.

2.9 Unanimous signs on three seeds are not power

Repeatedly, effects that were unanimous and large on a handful of creatures did not survive more of them. A smoothing sweep read +0.24 at $n = 3$ and +0.03 at $n = 6$; a salience-reward effect was +2.6 SE at $n = 12$ and null at $n = 16$; a vigilance score of 0.737 fell to 0.594 at 96 creatures. The project adopted a 3 SE bar at $n \leq 18$ as a direct consequence, and the bar must be fixed once and named, not chosen per test: one verdict printed “not yet a finding” for 2.9 SE against a 3 SE bar and then made a 2.7 SE contrast its headline because that test had been coded against 2.0.

3. Credit assignment and interference

Teaching a second, conflicting vowel target destroys the first. The surviving fraction is 0.22. This section is what the project established about why, and about what does not fix it.

3.1 The wipe has a mechanism, and it is not a representational collision

Two lessons interfere while driving disjoint neuron groups, so the groups are not what they share. What they share is the reward: a single broadcast scalar nudges the bias of every neuron in the module, so teaching B random-walks A's setting. Measured directly, B restores the very neurons A suppressed. Orthogonal targets are held simultaneously at 0.84 against the conflicting pair's 0.22, which locates the failure in the conflict rather than in capacity.

3.2 It is an axis collision, not a distance effect

A second lesson that is silent about the first lesson's axis is free; one that speaks to the same axis is destructive, confirmed at +11.5 SE. Distance between the two targets is not the variable.

3.3 A perfect per-neuron credit mask does not remove the wipe

An oracle mask — the host tells the creature which lesson is live, and the reward is confined to the half of the motor group that lesson uses — retains *less* than the broadcast rule it was meant to repair: $+0.109 \pm 0.057$ against $+0.320 \pm 0.052$, a deficit of -0.211 ± 0.074 , negative at every gain floor. It also learns less, gaining $+0.196 \pm 0.015$ against $+0.362 \pm 0.033$. This retracts an earlier claim of the opposite sign, which had rested on the estimator described in §2.8.

3.4 The mask's cost decomposes additively, and content and position cost the same

On end-of-teaching error, paired over 32 creatures:

Step	Contrast	Effect
masking at all	no mask → mask aimed correctly	+0.211
content	correct aim → random aim	+0.103
position	random half → central half	+0.102
total	central mask vs no mask	$+0.417 \pm 0.044$ (9.5 SE)

The two increments are equal to within 0.001, and each is about half the base cost of masking at all. The direct position contrast — a central mask against a randomly aimed one, at matched content and matched mask fraction — is $+0.102 \pm 0.009$, 11.5 SE.

The ordering on learning is: no mask, then correct aim, then random aim, then central. That a mask excluding *both* extremes of the group costs as much again as mis-aiming one is consistent with the leverage geometry of §4.

3.5 The aim of the mask during the interfering lesson recovers half of what masking costs

Two arms that mask the first lesson identically and differ only in which half they confine the reward to during the *second* lesson. Both therefore learn the first lesson equally well — verified by end-of-teaching error identical to $+0.0000 \pm 0.0000$ over 32 creatures — so the comparison is clean.

Gain floor	Aim correct	Aim wrong	Difference	n
0.00	+0.1092	-0.0873	-0.1965 ± 0.0510 (3.9 SE)	32

0.05	+0.1291	−0.0815	-0.2107 ± 0.0509 (4.1 SE) 29
0.10	+0.1327	−0.0782	-0.2109 ± 0.0508 (4.2 SE) 27
0.20	+0.0598	−0.0934	-0.1532 ± 0.0478 (3.2 SE) 18

Retention as a paired, jackknifed ratio of sums, swept over a floor on per-creature learning. The effect holds at every floor that retains enough creatures, which is the criterion the experiment’s own verdict gate applies.

Against a no-mask baseline of +0.320, a wrongly aimed mask sits −0.407 below it and correct aim recovers +0.197 — about 48%. The residual −0.211 is the part no amount of correct aim removes. So credit assignment here is not inert: what defeats the oracle is that the price of confining the reward exceeds what aiming it well buys back.

3.6 Position and content are not separable in a binary half-split

A structural result, and a negative one about the design rather than the creature. The mask is a contiguous half of the motor group and each lesson prefers exactly one half, so within a phase “the mask is in the wrong place” and “the mask selects the wrong neurons” are the same statement. No contrast built from end-aligned masks can separate them.

The only mask aligned with neither lesson — the central half, which leaves each lesson half of its preferred neurons on every trial — does separate them, and cannot answer the retention question for a second structural reason: it cuts learning by a factor of 29 (gain $+0.0125 \pm 0.0023$ against +0.362), so no creature clears even the lowest meaningful gain floor and retention has no denominator. **Varying position far enough to matter destroys the learning whose retention one would be measuring.** More creatures cannot fix this; it is not a power problem.

4. Leverage and the geometry of the motor pool

4.1 The block cost is leverage, not fraction

Freezing a quarter of the formant group costs 57% of the lesson and freezing a half costs 53%. The cost is flat in the fraction blocked, so there is no mask size that wins and nothing to gain from tuning one.

4.2 Leverage sits at one end

Freezing the lower half of the F1 group costs 9% of the lesson; freezing the upper half costs 53%. At single-neuron resolution one neuron costs 30% at the top of the group and 7% in the middle, and one cell of fourteen costs 30% while four cost 57%. An attempt to fit a leverage law to the single-neuron costs reproduced them to 1.68 percentage points and then failed the two half-block costs out of sample, so the asymmetry is measured and the law that tried to explain it is refuted.

4.3 The lesson silences the wrong side

Measured rather than fitted: a low formant target is reached by *suppressing* the upper half of the group, and a

high target by suppressing the lower half — mirror-image profiles, both suppression. An earlier account in which the lesson raised the half it needed is a decisive reversal.

4.4 The larynx reads nine scalars

The motor interface is nine scalar groups, and the object identity being taught sits at the null in all eighteen vocal measurements taken. Reordering the vocal population by its own response difference lifts discriminability, so the information is present in the slice and absent from the readout.

5. The voice and the body

5.1 The voice drones

There is no syllabic rhythm in the produced voice. A +9.99 dB modulation peak that indicated otherwise was a fit on the knee of a low-pass filter, not a spectral feature of the signal. Relatedly, a 3 Hz ring in the voice is the product of two filters — a fast rate constant and a gate smoothing term — and may be no oscillator at all; it scales threefold with self-hearing, and a deaf creature still rings at +3.1 SE, so self-audition is a gain on it rather than its cause.

5.2 The frame is a body part

A single finding that explains five separate refusals at once: the apparent syllabic frame is a resonant jaw at about 4.5 Hz, a property of the articulator rather than a central clock. Three independent routes from a frame to content are closed by it, and a jaw-gate mechanism that appeared to pass both halves of its test turned out to have a mis-set anchor, with the creature's own silences outperforming the imposed gate.

5.3 A blind controller beats feedback

A controller with no access to the sensed formant outperforms one with it: reach is immune to loop delay and a 905 ms loop is not. The precision advantage claimed for feedback was cleared by a sensorless controller, and an uninformative feature leaves only the intercept, which is the blind controller in disguise. Altered auditory feedback moves production by +0.01.

5.4 Intrinsic plasticity is the backlash

Formant actuator hysteresis measures 107 Hz with intrinsic plasticity enabled and 2.0 Hz with it off: the plant moves under its own controller. The backlash is per-neuron — pooling the regulator takes hysteresis from 107 Hz to 7 Hz and gives back 30 Hz of reach.

5.5 The ear hears its own formant and does not use it

The self-produced first formant is present in the auditory representation (+0.124 and +0.114 on two seed families) and unused by the motor system. The ear's estimate is well described by a fixed attenuation plus noise, with the attenuation free and the noise 56 Hz input-referred; the signal is localised, one frequency bin of 32 carrying 92% of the available benefit, which is destroyed by pooling. So the failure of closed-loop control here is a missing *mapping*, not a missing signal.

6. Representation and capacity

6.1 Vocabulary does not collapse, and nine words do not cost naming

Advantage over chance *rises* with vocabulary size rather than falling. At nine words against two six-word arrangements, total naming excess is conserved (-0.012 ± 0.019 and $+0.020 \pm 0.018$, opposite-sign nulls at $n = 36$). There is one real cost and it is not naming: delivered formant separation falls by 11.8 ± 3.5 Hz at matched trials per word, 3.3 SE. The shape of the word arrangement does not matter (-0.032 ± 0.027), so there is no preferred axis per creature.

6.2 The ceiling is an exponent

Delivered formant change scales with the aligned component of the learned bias as a power law with exponent about 0.61. Moving delivery by a factor requires moving the bias by a larger factor, which closes the naming line quantitatively rather than by argument: a commitment brake moved the aligned bias $1.07\times$ where $6.2\times$ was needed.

6.3 The derived index gap has no surviving cause

The creature derives its own context from an auditory rate average, and the resulting index is reliably *worse* when the teaching target tracks the heard word than when it is fixed, on 18 of 18 creatures — a gap of -0.056 ± 0.009 , 6.0 SE. Three candidate causes were tested and all three are dead: gate corruption has the wrong sign; silencing self-audition does not close the gap but *widens* it by 0.037 ± 0.015 , under a test with power to see a closure; and a drift account is excluded both by its own point prediction, 3.9 SE away, and by the learning arm decaying least. The gap is real and unexplained.

Two side results survived that line. Deafening improves the index in the control arm only ($+0.030 \pm 0.004$, 7.1 SE), and makes the tracking arm's partition more lopsided ($+0.066 \pm 0.018$, 3.7 SE). Separately, the derived context carrier is refuted directly: the index never separates the two lessons (0.23 against 0.23), while the auditory feature it is built from carries the word at 97.8% on held-out trials where the reward lands. The feature is fine; the extraction is not.

6.4 Two percent of a percent of the ticks does the work

0.2% of ticks account for the learning, and no schedule reproduces it at 3.3 SE over 32 creatures. The result carries no lesson about *which* ticks, which is stated here because it is the kind of finding that invites one.

7. Negative results that closed lines

Listed because the project's cost is dominated by them and because each removed a plausible mechanism from consideration.

Mechanism

Outcome

Oscillations as a rate-to-phase code

Built; spike-timing plasticity could not read it

Hippocampal pattern separation	Does not separate
Exuberant synaptogenesis	Costs 0.229; competition a further 0.051. Weight is not information
Dendritic error microcircuit	Settles and cancels nothing
Plateau-gated plasticity	Attenuates instead of selecting
Two-stage bandpass	Never appears on three seeds
Topographic tract	Oracle predicted 471 Hz; measured 1.00×
Region-shaped reward targets	A satisfied lesson still does not protect the first
Two-pool differential readout	Premise fails: only $\rho = 0.101$ of variance is shared
Adaptation current as a clock	Peak frequency follows its time constant at exponent 0.10 where 1.0 is required
Prototype freezing	Refused; it is the stability–plasticity dilemma
Replay / interleaving as shipped	Switching it off costs 0.3 SE
Reward that sees target distance	Worse, by 9.4 SE
Protocol axes (duration, blocking)	28× longer is no help; blocked teaching is harmful

8. Provenance

Every number above comes from a run of the project’s own experiment binary, gated before execution on --experiment verify reporting PASS with the pinned hash ad96f882becbee92 and a passing pinned-hash check. Runs that failed the gate are discarded rather than reported.

Sample sizes are creatures, not trials: a creature is one genome with one seed, grown and taught from birth. Where a paired statistic is quoted, creatures are matched by seed across arms, which is licensed by the per-job seed being a function of the repetition index alone and independent of the arm — verified by a shared baseline arm reproducing bit-identically across three separate multi-hour runs, and by a direct cross-run check of that arm against itself returning $+0.0000 \pm 0.0000$ over 32 creatures.

Retention figures are a ratio of sums with a leave-one-out jackknife standard error, swept over a floor on per-creature learning, and are reported only where the effect holds at every floor retaining at least three creatures in both arms. Section 2.8 explains why the more obvious estimator is not used.

Run logs, genome files and the experiment source are in the project repository. The findings here are recorded at greater length, with the reasoning that produced them and the criteria that would refuse them, in the project’s README.md and its memory store.

Appendix A. Excluded by the criterion

These project results are omitted from the body because a published result says or frames them first. They are listed so that the inclusion criterion of §1 can be checked.

Project result

External antecedent

Interference as a failure of contextual inference; a per-neuron mask should therefore help

Heald, Lengyel & Wolpert, COIN

Gating helps and is insufficient alone; stabilisation is the missing half

Masse, Grant & Freedman (2018)

Importance-weighted protection of earlier weights as the remedy

Kirkpatrick et al. (2017); Zenke, Poole & Ganguli (2017)

Node perturbation as the motor learning rule

Fiete, Fee & Seung

Sequence generation needs competition, not spike timing alone

Fiete, Senn, Wang & Hahnloser (2010)

Prototype freezing as a dilemma rather than a bug

Grossberg

A uniform cluster prior implicit in a competitive learning rule

MacQueen; Assran et al. (2023)

Ratio-of-sums estimation and the jackknife

Cochran; Quenouille; Tukey

Per-context reward prediction error having nothing to remove

Gadagkar et al. (2016)